

Design and Simulation of a 6T SRAM Cell Using LTspice

Project Report

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1. Abstract

This project implements a conventional 6-transistor (6T) Static Random Access Memory (SRAM) cell using LTspice. The work includes complete schematic design, CMOS device modelling, transient simulation, debugging of the circuit, and verification of five operating conditions: Hold, Write '1', Write '0', Read, and Read with Bit-Line Capacitance. The project demonstrates correct SRAM functionality and illustrates the effect of parasitic bit-line capacitance on read delay.

2. Introduction

SRAM is one of the most widely used memory technologies in modern processors because of its high speed and low latency. A single 6T SRAM cell stores one bit using two cross-coupled CMOS inverters. Unlike DRAM, SRAM does not require periodic refreshing, making it suitable for cache memories.

3. Objectives

- Design a complete 6T SRAM cell in LTspice.
- Verify Hold operation.
- Perform Write '1' and Write '0' operations.
- Verify Read operation.
- Study the influence of bit-line capacitance.

4. Circuit Description

The circuit consists of six MOSFETs. Two PMOS and two NMOS transistors form two cross-coupled CMOS inverters that store the data. Two additional NMOS access transistors connect the storage nodes (Q and QB) to complementary bit-lines (BL and BLB) whenever the Word Line (WL) is asserted. A 1 V supply is used together with Level-1 NMOS and PMOS SPICE models.

5. Design Methodology

The SRAM cell was first created using LTspice symbols. During implementation, power MOSFET symbols were replaced with monolithic CMOS devices to allow transistor sizing and custom model definitions. NMOS and PMOS LEVEL=1 models were added through SPICE directives. Initial conditions were used to define the stored logic state before transient analysis. Several debugging steps—including correcting transistor orientation, body connections, model names and access transistor connections—were required before obtaining correct operation.

6. Experimental Results

Experiment 1 – Hold: WL remained LOW and the stored data was retained without disturbance.

Experiment 2 – Write '1': BL=1 and BLB=0 while WL was HIGH. The cell successfully stored logic '1'.

Experiment 3 – Write '0': BL=0 and BLB=1 while WL was HIGH. The cell flipped to logic '0'.

Experiment 4 – Read: After precharging the bit-lines, WL was asserted and the stored value was read without destroying the stored data.

Experiment 5 – Read with Bit-Line Capacitance: Capacitors were connected between BL/BLB and ground to emulate practical parasitic capacitance. Larger capacitance increased the RC time constant, producing slower charging and discharging of the bit-lines.

7. Observations

- Cross-coupled inverters provide bistable storage.
- Access transistors isolate the cell during Hold operation.
- Write operations require complementary bit-line voltages.
- Read operation slightly perturbs the bit-lines but preserves stored data.
- Bit-line capacitance significantly affects read speed.

8. Challenges Faced

During the project several practical LTspice issues were solved, including MOSFET model errors, unsupported device symbols, transistor orientation, source/drain assignment, custom SPICE directives, and verification of expected transient waveforms. Solving these issues improved understanding of both SRAM design and circuit simulation.

9. Conclusion

The project successfully demonstrated all major operating modes of a conventional 6T SRAM cell. The simulations confirmed proper hold, write, and read functionality, while the capacitive loading experiment highlighted the practical impact of bit-line parasitics on memory performance.

10. Future Work

Future improvements include using BSIM transistor models, measuring read/write delay and static noise margin (SNM), estimating power consumption, and extending the design to an SRAM array with peripheral circuits such as precharge, sense amplifiers, and row decoders.

References

1. Neil Weste and David Harris, CMOS VLSI Design.
2. Jan M. Rabaey, Digital Integrated Circuits.
3. LTspice XVII Documentation.